

Mathematics and Terminology — every formula used

Notation, then each method derived from its definition, then a worked example with real numbers from this project. Nothing here is assumed known beyond matrix multiplication and the idea of a variance.

0 * Notation

symbol	means	in this project
N	number of items	9,533 image-caption pairs; 1,000 held out
d_A, d_B	width of a space	MobileCLIP 512, SigLIP 768, DINOv2-L 2048
$X \in \mathbb{R}^{(N \times d)}$	one space's vectors, one row per item	row i of every array is the SAME item in every space
x_i	row i of X	the vector for item i
W	a fitted linear map	the adapter; the hub entry maps
$\ x\ $	length of x, = $\sqrt{x \cdot x}$	after L2 normalisation, $\ x\ = 1$
X^T	transpose	swaps rows and columns
tr	train rows	8,533 of the 9,533
k	neighbourhood size	1 ... 500 in the scale sweep

1 * Cosine similarity — the base measure

Every comparison in the project reduces to angles between vectors, not distances. Length is discarded first.

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L2 normalisation    x_hat = x / ||x||    ->    every vector has length 1

cosine(x, y) = (x * y) / (||x|| ||y||) = x_hat * y_hat

after normalisation the dot product IS the cosine – one multiplication, no division

range: +1 identical direction * 0 orthogonal * -1 opposite
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Why cosine and not Euclidean distance. For unit vectors the two are equivalent — $\|x_{\text{hat}} - y_{\text{hat}}\|^2 = 2 - 2(x_{\text{hat}} \cdot y_{\text{hat}})$ — so cosine is just a monotone rescaling of distance, and it is the quantity a vector index actually computes.

The trap. Cosine says nothing about whether the space is healthy. In GPT-2's caption space every random pair reads +0.999, so a cosine of 1.000 between a prediction and its target means almost nothing. Always read cosine against the space's own baseline (section 8).

2 * Ridge regression — the map, in closed form

The central object of the project. Given paired rows of two spaces, find the matrix that best sends one to the other.

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PROBLEM    find W minimising    ||X W - Y||^2_F + alpha ||W||^2_F
           (fit error)          (penalty on size)

SET the derivative to zero:
    2X^T(XW - Y) + 2alpha W = 0
    (X^T X + alpha I) W = X^T Y

SOLUTION    W = (X^T X + alpha I)^-1 X^T Y

one linear solve. No gradients, no epochs, no initialisation, no random seed.
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What alpha does. With $\alpha = 0$ this is ordinary least squares, which becomes unstable when $X^T X$ is near-singular — common when d is large relative to N. Adding α lifts every eigenvalue by α , guaranteeing invertibility and shrinking the solution toward zero. In this project α is chosen by a sweep on training-internal validation, never on the evaluation split.

Why closed form matters for the claim. A trained mapper could succeed by learning the task itself, which would prove nothing about convergence. A closed-form solve has no capacity to hide behind: it is the best linear map that exists for this data, found exactly.

quantity	formula	this project
held-out R^2	$1 - \ Y - XW\ ^2 / \ Y - \text{mean}(Y)\ ^2$	0.592 (Experiment B)
mean cosine	mean over i of $\cos(x_i W, y_i)$	0.900

quantity	formula	this project
rows per dimension	$N_{\text{train}} / d_{\text{input}}$	5.9–9.8 everywhere; floor of 5 pre-registered

3 * Procrustes — the rotation-only map

The same problem as ridge, but W is forced to be a pure rotation or reflection. Comparing the two says whether the spaces differ only in orientation.

CONSTRAINT $R^T R = I$ (orthogonal: preserves all lengths and angles)

PROBLEM minimise $\|X R - Y\|_F^2$ subject to $R^T R = I$

SOLUTION take the SVD of the cross-product:

$$X^T Y = U S V^T \rightarrow R = U V^T$$

the S is DISCARDED — that is exactly what makes it rotation-only

Reading the comparison. Procrustes is a strictly harder problem than ridge, so its score can only be lower. The informative quantity is the GAP.

result	meaning
Procrustes $\approx$ ridge	the spaces are the same shape, just turned — near-isometric
Procrustes $\ll$ ridge	the spaces differ in STRETCH along different directions (anisotropy), not merely in orientation

This project: 0.038 against ridge 0.592 — a 15× gap. The map's own singular values span 69×, meaning it stretches some directions 69 times more than others. Hence Finding 2, *related but not rigid*.

4 * PCA and whitening — how the hub is built

The hub is not trained. It is one SVD and one rescaling.

STEP 1 standardise each encoder's block, then concatenate

$$Z = [(X_1 - \mu_1) / \sigma_1 \mid (X_2 - \mu_2) / \sigma_2 \mid \dots] \quad (8,533 \times 7,680 \text{ here})$$

STEP 2 centre on the TRAIN rows only

$$Z_c = Z - \text{mean}(Z_{\text{train}})$$

STEP 3 SVD of the centred training block

$$Z_c = U * \text{diag}(S) * V^T$$

V^T rows = principal DIRECTIONS (orthogonal unit vectors)

S = how far the data STRETCHES along each, sorted descending

STEP 4 singular value $\rightarrow$ variance

$$\text{covariance} = Z_c^T Z_c / n = V * \text{diag}(S^2/n) * V^T$$

$$\text{so } \lambda_k = S[k]^2/n \quad \text{and} \quad \text{std along } k = S[k]/\text{sqrtn}$$

STEP 5 whiten: divide each kept direction by its own std

$$W_{\text{hub}} = V[:,d]^T / (S[:,d]/\text{sqrt}(n))$$

every kept direction now has variance 1 — no direction is 'bigger' than another

The sqrtn. It appears because covariance divides by the row count while the SVD does not. It is the only step that is not obvious from the shapes.

Worked, with this project's numbers ($n = 8,533$ train rows, concatenated width 7,680):

hub width d	$S[d]$	$\lambda = S^2/n$	$\text{sqrt}\lambda = S/\text{sqrtn}$	amplification $1/\text{sqrt}\lambda$
64	376.82	16.64	4.079	0.25
128	260.42	7.95	2.819	0.35
256	181.61	3.87	1.966	0.51
512	124.49	1.82	1.348	0.74
768	98.92	1.15	1.071	0.93

Check the $d = 512$ row by hand: $\text{sqrt}(8533) / 124.49 = 92.37 / 124.49 = 0.742$.

Why the amplification column matters. S falls as d grows, so $1/\text{sqrt}\lambda$ rises: later directions carry little variance and whitening scales them UP to match the strong ones. That is the proposed mechanism for the hub width cliff — and testing it is how it came to be falsified, since the rise is smooth while the cliff is a discontinuity.

5 * Retrieval metrics

similarity matrix $S = P G^T$ (predictions $\times$ gallery, both L2-normalised)

$R@k$ = fraction of queries whose TRUE match is in the top k of its row

chance = k / N_{gallery} $\rightarrow$ with a 1,000-item gallery, $R@1$ chance = 0.001

% of ceiling = $R@k(\text{system}) / R@k(\text{a jointly-trained model, SAME protocol})$

The trap. $R@1$ is meaningless without the gallery size. 0.967 among 10 candidates and 0.557 among 1,000 are not comparable. Every retrieval number in this project uses a 1,000-item gallery so the figures can be compared across experiments.

6 * AUC — and why it is not one number

$AUC = P(\text{score(a true pair)} > \text{score(a false pair)})$

equivalently, the area under the true-positive vs false-positive curve as the threshold sweeps

0.5 = chance * 1.0 = perfect separation

IT DEPENDS ENTIRELY ON WHICH FALSE PAIRS YOU CHOOSE.

negative used	what it asks	Experiment B
a RANDOM other item	easy — usually semantically unrelated	AUC 0.999
the HARDEST of 1,000	the closest wrong candidate; an extreme-value draw	AUC 0.848
all 1,000, pick one	$R@1$ — strictly the hardest	0.987
no adapter at all	the control	AUC 0.477 = chance

The spread is the evidence against overfitting. A memorised map does not drop 0.15 when the negative gets harder. Difficulty of the negative set, not memorisation, explains the range.

7 * Neighbourhood overlap — the local measure

$N_A^k(i)$ = the k nearest items to i , inside space A

$\text{overlap}_k(i) = |N_A^k(i) \cap N_B^k(i)| / k$

chance = k / N $\rightarrow$ report the RATIO: $\text{overlap} / (k/N)$

NO MAP IS FITTED. This is why it is evidence that structure exists BEFORE any fit.

The ceiling trap, which cost this project a correction. The ratio is bounded above by N/k — 1,500 at $k=1$ but only 3 at $k=500$. A pair with perfect agreement at every scale would still show a falling ratio. The chance-corrected form is comparable across scales:

$\text{kappa} = (\text{observed} - \text{chance}) / (1 - \text{chance})$ bounded $[0,1]$ at EVERY k

measured here: rises to a peak near $k = 20-50$, then falls — NOT the monotone collapse the raw ratio suggested. See Appendix C.13.9.

COORDINATE FRAME (the axes cosine is read against) · §8 and C.11

Same points, different rulers. Two encoders can place items in the SAME relative arrangement (shape) while using unrelated axes · which is why raw cross-space cosine sits at ~ 0 even when shapes agree. Cosine reads content best when the frame spreads meaning across many directions (isotropy, §8). A collapsed frame packs everything on one axis (GPT-2, pair-cos +0.999): items look alike, cosine can't separate them. Whitening installs a good frame for free: GPT-2 $R@1$ 0.130 $\rightarrow$ 0.478 (C.11). A target need not be perfectly isotropic · bge-m3 is anisotropic (pair-cos +0.57) yet writable at 98.1%, because a map is fitted INTO it, so the writability test (not pair-cosine) is what qualifies a target. SBERT has the cleaner frame (+0.135) and scores marginally better; bge is kept for comparability, not merit (C.13.4).

8 * Geometry diagnostics

Two numbers that predict how a space will behave before any map is fitted.

EFFECTIVE RANK (spectral entropy – 'how many directions matter')

eigenvalues of the covariance: $\lambda_1 \dots \lambda_d$
 $p_k = \lambda_k / \sum \lambda$ (normalise to a probability distribution)
 $\text{eff.rank} = \exp(-\sum p_k \cdot \log p_k)$

all variance in one direction $\rightarrow 1$
variance spread evenly over $d \rightarrow d$

MEAN PAIR-COSINE (how alike are two RANDOM items)

mean over $i \neq j$ of $\cos(x_i, x_j)$
 ~ 0 healthy * $+0.999$ collapsed
width-INDEPENDENT, which is why it is the preferred flag

space	eff. rank / width	pair-cos	reading
img_small	252 / 768 = 33%	+0.060	healthy
txt_sberr	93 / 768 = 12%	+0.136	healthy — despite the low ratio
txt_bge	98 / 1024 = 10%	+0.570	degenerate
txt_gpt2	6.5 / 768 = 0.8%	+0.999	collapsed — every item on top of every other

Why pair-cosine and not the rank ratio. An earlier version of this analysis flagged spaces by eff.rank/width and misclassified SBERT, because text encoders legitimately carry lower ratios than image encoders. The ratio conflates modality with collapse; pair-cosine does not.

9 * Structural similarity measures

CKA (Centered Kernel Alignment – global structure)

$K = XX^T$, $L = Y Y^T$, both double-centred

$\text{CKA} = ||X^T Y||^2_F / (||X^T X||_F * ||Y^T Y||_F)$

invariant to rotation, isotropic scaling, permutation

CKNNA (the same, restricted to MUTUAL k-nearest neighbours)

as $k \rightarrow n$ it recovers plain CKA – one knob spanning local to global

CCA (shared vs private directions)

find a, b maximising $\text{corr}(Xa, Yb)$; repeat orthogonally

the spectrum of correlations = how many directions are genuinely shared

The width confound. CKA has a positive null that rises with embedding width: independent spaces of 2048 dimensions score higher than independent spaces of 768. The fix is a permutation null — shuffle one side's rows, preserving widths and geometry while destroying the correspondence, and subtract.

measured nulls in this project

k-NN overlap null = $+0.016$ = exactly k/N $\rightarrow$ no correction needed

distance rank rho null = $+0.000$ $\rightarrow$ no correction needed

CKA null = $+0.094$, rising with width $\rightarrow$ CORRECTION REQUIRED

10 * Gromov-Wasserstein — matching without pairs

Every method above needs to know which row corresponds to which. This one does not.

given only the two spaces' INTERNAL distance matrices D_A , D_B

find a soft assignment T (rows sum to $1/N$, columns too) minimising

$$\sum_{ijkl} |D_A[i,k] - D_B[j,l]|^2 * T[i,j] * T[k,l]$$

in words: if i and k are far apart in A , then their assigned partners j and l should be far apart in B . Only SHAPE is used — never coordinates.

solved with an entropic regulariser ϵ (swept, never tuned on eval)

Result: 39.7% of items matched exactly = 595× chance, with zero paired examples. The correspondence partially identifies itself from geometry alone.

11 * Intrinsic dimension (TwoNN)

for each point, take its two nearest-neighbour distances $r_1 < r_2$

$$\mu = r_2 / r_1$$

on a d -dimensional manifold, μ follows a Pareto law with exponent d , so d is recovered from the slope of $\log \mu$ against $-\log(1 - F(\mu))$

measured here: 11.2 – 19.9 inside spaces 768–2048 wide (<3% of ambient)

12 * Statistics used for honesty

quantity	formula	used for
standard error of an R@1 difference	$\sqrt{2p(1-p)/n}$	± 0.022 on 1,000 queries — the reason a +0.008 residual is called noise
ratio to chance	$\text{observed} / (k/N)$	every neighbourhood figure
chance-corrected agreement	$(\text{obs} - \text{chance}) / (1 - \text{chance})$	comparing overlap across different k
z against a permutation null	$(\text{obs} - \text{mean_null}) / \text{sd_null}$	the width-confound calibration
Spearman rho	Pearson correlation of RANKS	shape agreement; robust to monotone warps, so collapse does not depress it
bootstrap CI	resample queries, recompute, take percentiles	F5's +0.0244 data effect

THE WHOLE PROJECT IN FOUR FORMULAS

1. the map $W = (X^T X + \alpha I)^{-1} X^T Y$
2. the hub $W_{\text{hub}} = V[:,d]^T / (S[:,d] / \sqrt{n})$
3. the test $R@k$ against chance k/N and against a matched ceiling
4. the check the same quantity under a permutation null

Everything else is a diagnostic that tells you when one of these four is lying.

8b · Three distinctions this project turns on

Concepts that came up repeatedly and decide how a number is read.

CENTRED vs UNCENTRED spectrum · centre for PCA, do NOT centre for collapse detection

Centring subtracts the mean vector before the SVD, removing exactly one direction · the one every item shares. For PCA (§4) that is correct: a constant offset is not a principal component. For COLLAPSE detection (§8) it is fatal: the shared direction IS the collapse, so a centred effective rank scores a collapsed space the same as a healthy one, and cannot reproduce GPT-2's published 6.5. Rule: effective rank for anisotropy is computed UNCENTRED.

verified: one-dominant-direction space -> centred eff.rank 594, uncentred 1.0

ANGLE and ORTHOGONAL · cosine is literally the cosine of the angle between two vectors

cos +1 -> 0 deg (same) cos 0 -> 90 deg (ORTHOGONAL) cos -1 -> 180 deg (opposite)

deg (

Two RANDOM high-dimensional vectors are almost always near-orthogonal · mean cosine 0, and the spread shrinks as dimension grows (about 0.036 at 768d). This is why raw cross-space cosine near 0 is a real finding, not a triviality: alignment must be PUT there by training, because in high dimensions accidents are perpendicular. It is also why a random-map control lands at chance (0.001) · a random matrix maps into essentially orthogonal directions.

READABLE vs WRITABLE · a space has two directional roles, and they are not symmetric

READABLE = a linear map can pull content OUT of a space (use it as a SOURCE); governed by whether the content is recoverable · effective rank. WRITABLE = once you predict a vector back INTO the space, cosine can RANK items there (use it as a TARGET); governed by whether items are separable · pair-cosine.

space / role			
bge-m3 (target)	98.1%	98.1%	symmetric, healthy
GPT-2	95.1%	24.8%	rank 6.6 recoverable, but pair-cos +0.999

They diverge because reading needs recoverable content and writing needs separable items · different properties. A read-out TARGET must be writable, which is why the target test uses writability (bge 98.1%, SBERT 101.0%), not pair-cosine, and why GPT-2 is a hub member yet could never be the target.

PERCENT OF NATIVE (% of native head) · the hub's central ratio metric

% of native = R@1_transfer / R@1_native_head x 100

Denominator: a head fitted specifically for THAT encoder on the same eval set. **Numerator:** the FROZEN hub head · trained only on DINOv2-small coordinates · applied unchanged to the new encoder's hub projection.

Why a falling % is NOT falling performance. Absolute R@1 RISES with encoder quality (0.398 / 0.442 / 0.468 across the DINOv2 ladder), but the denominator (the native head's R@1) rises faster · so the percentage FALLS (111.5 / 95.9 / 92.9%). A 92.9% for DINOv2-large does not mean it performs worse than DINOv2-small's 111.5%; it means its native head is a harder ceiling to match. The percentage should always be read alongside the raw R@1.

See report E.13 and Table C12 for the full calculation per encoder.